Fact and Dimension Identification

DailyBasket Data Warehouse - Phase 1
Hamza Abbas (L24-2500) and Sadeem Arshad (L24-2502)

1. Business Processes and Fact Table Types

Business Process	Fact Table	Fact Type	Why this type
Sales Transactions	fact_sales	Transaction fact	Each sale is one event that happens once.
Inventory Snapshots	fact_inventory_snapshot	Periodic snapshot fact	Stock is measured at regular times (end of each day).
Promotion Events	fact_promotion_event	Factless fact	We only record that a product was in a promotion. There is nothing to measure.

2. Fact Tables and Measures

fact_sales (Transaction fact)

Grain: One row represents one product purchased in one order transaction (one order line).

Measure	Meaning	Additivity
quantity	Units bought	Additive
unit_price	Price of one unit at sale time	Non-additive
discount_amount	Discount on the line	Additive
gross_amount	quantity x unit_price	Additive
net_revenue	gross_amount minus discount_amount	Additive
cost_amount	quantity x unit cost	Additive
profit_amount	net_revenue minus cost_amount	Additive

Degenerate dimensions: order_id, order_line_number (kept in the fact table, no separate dimension).

fact_inventory_snapshot (Periodic snapshot)

Grain: One row represents the end-of-day stock level of one product in one region.

Measure	Meaning	Additivity
quantity_on_hand	Units in stock at end of day	Semi-additive (do not add across dates)
quantity_received	Units received that day	Additive
quantity_sold	Units sold that day	Additive
reorder_level	Minimum stock before re-ordering	Non-additive
stock_value	quantity_on_hand x unit cost	Semi-additive (do not add across dates)

fact_promotion_event (Factless fact)

Grain: One row represents one product being part of one promotion on one active day.

No measures. We answer questions by counting rows (COUNT(*)). The three foreign keys together form the primary key.

3. Dimensions

Dimension	What it describes	Main attributes	Hierarchy
dim_date	When	full_date, day_name, month_name, quarter, year, is_weekend	Year > Quarter > Month > Day
dim_customer	Who bought	customer_name, gender, age_group, customer_segment, signup_date	Segment > Customer
dim_product	What was sold	product_name, brand, subcategory, category, unit_cost, list_price	Category > Subcategory > Product
dim_region	Where	city, province, country, region_zone	Country > Province > City
dim_channel	How it was sold	channel_name, channel_type	Channel Type > Channel
dim_promotion	Which offer	promotion_name, promotion_type, discount_percent, start_date, end_date	Promotion Type > Promotion

4. Natural and Surrogate Keys

Table	Natural Key (from source)	Surrogate Key (warehouse)
dim_date	full_date	date_key (YYYYMMDD)
dim_customer	customer_id	customer_key
dim_product	product_id	product_key
dim_region	region_id	region_key
dim_channel	channel_id	channel_key
dim_promotion	promotion_id	promotion_key (0 = No Promotion)
fact_sales	order_id + order_line_number	sales_key
fact_inventory_snapshot	date + product + region	inventory_snapshot_key
fact_promotion_event	date + product + promotion	composite of the 3 foreign keys

Surrogate keys are simple numbers made by the warehouse. They stay the same even if the source system changes its IDs.

5. Bus Matrix (which process uses which dimension)

Business Process	Date	Customer	Product	Region	Channel	Promotion
Sales Transactions	X	X	X	X	X	X
Inventory Snapshots	X		X	X		
Promotion Events	X		X			X

dim_date, dim_product, dim_region and dim_promotion are shared (conformed) dimensions: the same table is used by more than one fact table.

6. Schema Diagram

DailyBasket Data Warehouse - Star Schema (Fact Constellation)



GRAIN fact_sales: One row represents one product purchased in one order transaction (one order line).

GRAIN fact_inventory_snapshot: One row represents the end-of-day stock level of one product in one region.

GRAIN fact_promotion_event: One row represents one product being part of one promotion on one active day. (factless: no measures)

PK = Primary Key FK = Foreign Key

NK = Natural Key DD = Degenerate Dimension

M = Fact Measure 1 --- N = one dimension row, many fact rows